

Evidence-Condition Geometry for Completion Decisions in Dynamic Agent Workflows

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Abstract

Dynamic agent workflows often need to decide whether a task is complete before the size and structure of the remaining workload are known. Common stop signals, including no-new findings, agent agreement, stable summaries, and self-reported completion, are useful local evidence, but they do not specify the conditions under which that evidence was produced. This paper studies the resulting certificate mismatch: a workflow can use evidence gathered under a localized source-route condition to support a global completion claim. We introduce source-route evidence-condition geometry as a runtime diagnostic for this failure mode and define a conservative controller that rejects SAFE when the evidence condition is too localized, repairs weak plausible strata, and returns SAFE, CONTINUE, or ABSTAIN. Across four bounded completion-audit tasks, homogeneous route reuse produces localized evidence conditions and would lead to false certification if accepted as a global stop. Broader source-route exposure improves completion eligibility, while an external `urllib3` audit shows the key boundary: broad exposure is not sufficient for SAFE when residual evidence remains. The supported claim is therefore restrained: local evidence cannot automatically certify global completion; completion decisions require the evidence condition itself to be audited.

1 Introduction

Agent workflows are increasingly used for open-ended tasks whose true workload is not known at runtime: searching a repository for all instances of a behavior, auditing a document set for policy exceptions, checking whether an implementation covers relevant failure modes, or deciding whether a multi-agent process has exhausted useful evidence. Such workflows must eventually stop. The difficulty is that the absence of newly found evidence is not the same object as evidence that the intended workload is complete.

The failure studied in this paper is *certificate mismatch*. A workflow may produce a plausible local stop signal: repeated scans return no new findings, agents agree with each other, subtasks appear closed, or a summary becomes stable. These signals are conditioned on the parts of the task that were searched and on the routes by which they were searched. If the workflow has repeatedly applied the same route to the same subset of sources, a no-new signal may certify only local exhaustion. It does not automatically support a global claim

that the task is complete over its intended scope.

This distinction is important because many dynamic workflows operate under hidden total workload. The system does not know how many relevant items, counterexamples, or unresolved cases exist. More importantly, it may not know which source-route conditions are necessary for a stop claim to be meaningful. Auditing a file for timeout behavior, for example, does not certify that the same file has been audited for TLS behavior, exception behavior, retry behavior, or cleanup behavior. The same source can support different kinds of evidence depending on the route applied to it.

We formalize this observation with *evidence-condition geometry*. The basic unit is a source-route stratum, consisting of a bounded evidence region and an audit or search route. At runtime, the workflow accumulates exposure counts over these strata. The normalized exposure distribution describes the condition under which completion evidence was produced. When this distribution is localized, stop evidence is locally conditioned. It may be useful, but it is not a global completion certificate.

Based on this geometry, we define an evidence-condition controller. At a proposed stop, the controller computes the source-route exposure condition, checks whether the condition is broad enough to make the completion claim eligible for certification, and triggers repair over weak plausible strata when necessary. The controller is intentionally conservative: broad exposure gives completion eligibility, not automatic safety. A SAFE decision also requires that repair or audit reveal no residual evidence. If residual evidence appears, the controller outputs CONTINUE; if budget is exhausted without a defensible certificate, it outputs ABSTAIN.

Our experiments evaluate this controller on four bounded completion-audit task families: two generated scored tasks and two external real-repository audits with pattern-defined oracles. These tasks are not presented as a benchmark for the best item discovery policy. Instead, they test whether source-route exposure localization diagnoses unsafe stop claims and whether a controller can avoid accepting locally conditioned evidence as global completion proof.

This paper makes four contributions. First, it formulates false completion as a certificate mismatch between the scope of a stop claim and the source-route condition under which supporting evidence was produced. Second, it introduces source-route exposure geometry as a runtime-computable diagnostic for whether completion evidence is locally or broadly conditioned. Third, it defines an evidence-condition controller that outputs SAFE, CONTINUE, or ABSTAIN by combining

exposure breadth with residual repair evidence. Fourth, it provides controlled and external validation showing that homogeneous route reuse creates false certification risk, while broader source-route evidence improves completion eligibility without being sufficient for safety.

2 Related Work

Technology-assisted review and total recall. Technology-assisted review and total-recall methods study how to retrieve nearly all relevant items under labeling and review budgets [Grossman and Cormack, 2011, Cormack and Grossman, 2014, 2016]. This literature is adjacent because our scored subclasses report recall where a bounded oracle exists. The target decision in this paper is different. We do not primarily optimize an item retrieval policy. We ask whether a dynamic workflow has enough conditioned evidence to certify a completion claim. Item discovery is therefore an evaluation subclass, not the definition of the problem.

Active search and active learning. Active search chooses actions to find positives efficiently, while active learning chooses examples to improve a model [Lewis and Gale, 1994, Settles, 2009, Garnett et al., 2012]. Our residual-potential repair instance resembles active search in that it prioritizes strata that may produce new evidence. Its role is narrower: it is used inside a stop controller to test whether a proposed completion certificate has weak source-route regions that still contain residual evidence. The controller’s objective is not only yield; it must also decide whether the evidence supporting the completion claim was gathered under conditions broad enough for that claim.

Missing mass and unseen evidence. Missing-mass and unseen-species methods estimate how much probability mass remains unseen after sampling [Good, 1953, Chao, 1984]. The motivation is close to false stopping under unknown workload, but our object is not a scalar estimate of hidden mass. The object is the runtime evidence condition. A workflow can have many observations and still be localized if those observations come from a small subset of source-route strata.

Multi-agent agreement and verification. Work on ensembles, self-consistency, and verifier-guided systems studies how multiple predictions or checks can improve reliability [Dietterich, 2000, Cobbe et al., 2021, Wang et al., 2023]. Our diagnostic is not semantic similarity between agent outputs. It records where and how the workflow searched. Agents may agree because the task is complete, or because they repeatedly examined the same local region through the same route. Source-route exposure geometry makes that distinction observable at runtime.

3 Problem Setting

We study workload-unknown dynamic agent workflows. A workflow receives a task, allocates work across agents or tools, gathers evidence under a finite budget, and must decide whether the task is complete. Let

$$W = (T, S, R, A, B),$$

where T is the task, S is a set of sources, R is a set of routes, A is the set of agents or tools, and B is a finite budget. A source is a bounded evidence region, such as a file, module, document, source family, or repository component. A route is an audit or search lens applied to a source, such as a timeout audit, TLS audit, exception audit, policy-exception audit, or cleanup audit.

At runtime, the workflow produces evidence events

$$e_t = (a_t, x_t, r_t, u_t, o_t, c_t),$$

where a_t is the agent or tool, $x_t \in S$ is the source, $r_t \in R$ is the route, u_t is the action, o_t is the observation, and c_t is the cost. An event may be a search, scan, extraction, verification, summary, or stop proposal. The key point is that evidence is conditioned: each event is produced under a source-route condition.

A source-route stratum is

$$s = (x, r), \quad x \in S, r \in R,$$

and the intended evidence-condition space is $\Omega = S \times R$. A stop claim is a workflow assertion

C_t : T is complete over the intended workload scope.

Signals such as no-new findings, agent agreement, stable summaries, or exhausted assigned subtasks are evidence for C_t only under the source-route conditions that produced them.

A false certification occurs when the workflow accepts SAFE for a stop claim that is not supported by the bounded evaluation oracle:

$$\text{SAFE}(C_t) = \text{true}, \quad \text{complete}(T) = \text{false}.$$

In item-discovery subclasses, we operationalize this as accepting SAFE while recall is below the chosen completion threshold. In broader completion-audit settings, false certification means accepting a global completion claim while relevant support, contradiction, or unresolved evidence remains outside the evidence condition.

4 Evidence-Condition Geometry

Let $v_t(s)$ be the runtime-visible exposure count for source-route stratum s up to time t . Exposure can be visits, scans, tool calls, route-specific audits, or other logged search actions. Define

$$p_{\text{exp}}(t, s) = \frac{v_t(s)}{\sum_{s' \in \Omega} v_t(s')}.$$

This distribution is a point on the source-route coverage simplex. It describes the condition under which completion evidence was produced.

When $p_{\text{exp}}(t, s)$ is concentrated on a small subset of strata, no-new or agreement evidence is locally conditioned. It may certify local exhaustion, but it does not certify global completion. We summarize this distribution with two operational quantities:

$$\text{support}(t) = \frac{|\{s : v_t(s) > 0\}|}{|\Omega|},$$

$$G_{\text{exp}}(t) = \text{Gini}(\{v_t(s) : s \in \Omega\}).$$

These are summaries of the exposure distribution, not additional core variables. They are used to decide whether the evidence condition is broad enough for the stop claim.

Figure 1 shows the certificate mismatch mechanism. Figure 2 shows the source-route geometry and the distinction between localized, broad-but-still-productive, and broad-stable evidence conditions. Source-only geometry is too coarse because it collapses routes inside a source. Source-route-action geometry is more detailed, but it is sensitive to logging conventions and the current evidence does not show a stable advantage over source-route. We therefore use source-route as the default geometry because it captures both where and how evidence was produced while remaining interpretable and runtime-computable.

5 Evidence-Condition Controller

At a proposed stop, the controller checks whether the current evidence condition can support the stop claim. It first computes $v_t(s)$, $p_{\text{exp}}(t, s)$, $\text{support}(t)$, and $G_{\text{exp}}(t)$. If the evidence condition is too localized, it rejects SAFE and triggers repair. A repair rule selects weak but plausible source-route strata using only runtime-visible information. After repair, the controller outputs CONTINUE if new residual evidence appears. It outputs SAFE only if the evidence condition is broad and weak plausible gaps are absent. If budget is exhausted and the certificate remains indefensible, it outputs ABSTAIN.

The controller is conservative by design. It does not infer completion merely from confidence, agreement, or no-new evidence. It asks whether the evidence condition is strong enough for the scope of the stop claim. In the current implementation, support and Gini thresholds instantiate an operational eligibility check. These thresholds are not claimed as universal laws; the invariant is the controller structure that separates eligibility from SAFE.

Residual-potential repair. The repair instance used in the experiments is

$$\text{priority}(s) = \text{under_exposure}(s) \cdot \text{runtime_potential}(s).$$

The under-exposure term gives priority to weak parts of the current completion certificate. The runtime-potential term uses visible information such as source text, route names, source

size, route match counts, or non-oracle ledger signals. It is not allowed to use oracle labels, oracle totals, oracle missing mass, undiscovered true item counts, post-hoc recall, or scorer-visible target distributions. Residual-potential is a mechanism-aligned repair instance. We do not claim it is optimal or generally superior to all active-search alternatives.

6 Experimental Protocol

Tasks. We evaluate on four task families. `policy_docset_v1` is a generated bounded policy document task. `code_repo_v1` is a generated bounded code repository task. `requests` is an external real-repository audit over a local snapshot of the Python `requests` package. `urllib3` is a second external real-repository audit. The generated tasks are controlled scored subclasses. The external tasks use real source structure but pattern-defined oracles, so they are stronger than toy tasks and weaker than fully human-annotated completion-audit benchmarks.

Conditions. We compare homogeneous route reuse, route-partitioned audit, and, where available, extended or near-complete audit. Homogeneous conditions test whether repeated local evidence can produce unsafe stop claims. Route-partitioned and extended conditions test whether broader source-route exposure improves completion eligibility and whether the controller can distinguish eligibility from safety.

Repair challengers. We compare random, low-exposure, high-potential, residual-potential, and free-search continuation challengers. These challengers are evaluated as evidence-condition repair mechanisms, not as standalone item-finding algorithms.

Metrics and leakage control. Primary metrics are false certification, controller decision, support ratio, exposure Gini, repair gain, cost-normalized evidence, and overlap between high-potential and residual-potential. Recall is reported for scored subclasses, but recall is not observed by the controller. Oracle labels are used only after trajectories and challenger choices are fixed to score whether a stop would have been false.

7 Results

7.1 Localized Evidence Creates False Certification Risk

Table 1 reports the main diagnostic. Across all four tasks, homogeneous route reuse produces localized evidence conditions and would cause false certification if accepted as SAFE. Base support ratios range from 0.200 to 0.333, and base exposure Gini values are high: 0.750 on `code_repo_v1`, 0.771 on `policy_docset_v1`, 0.889 on `requests`, and 0.915 on `urllib3`. Under the bounded oracles, the corresponding

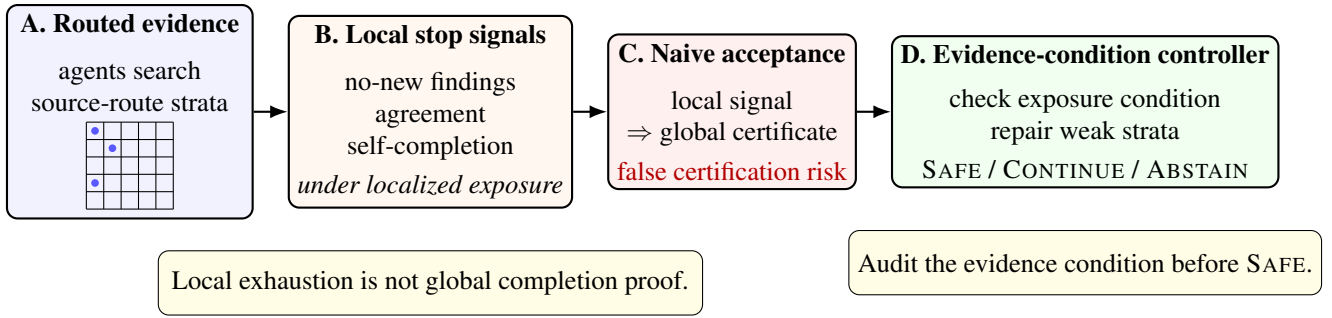


Figure 1: Certificate mismatch in completion decisions. Dynamic workflows produce stop evidence under specific source-route conditions. Repeated no-new findings, agent agreement, or stable summaries can indicate local exhaustion, but they do not automatically support a global completion claim. The evidence-condition controller checks whether the exposure condition is broad enough for the claim, repairs weak plausible strata, and returns SAFE, CONTINUE, or ABSTAIN.

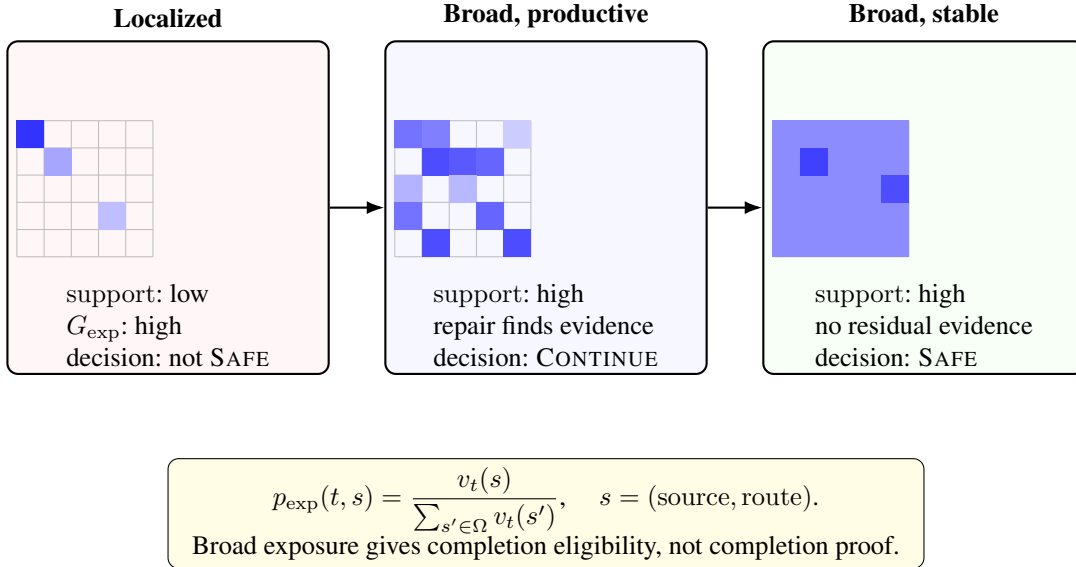


Figure 2: Source-route exposure geometry. The runtime exposure distribution over source-route strata records where and how completion evidence was produced. Homogeneous route reuse concentrates exposure on a small part of the coverage simplex, so stop evidence is locally conditioned. Broader route-partitioned exposure makes a stop claim eligible for certification, but SAFE additionally requires that repair or audit reveal no residual evidence.

base recalls are below the completion threshold in all four tasks.

The controller rejects these local stops. This is the main diagnostic result: the issue is not simply that an agent failed to find items, but that the stop evidence was produced under a narrow source-route condition and therefore could not support a global completion claim.

7.2 Broad Exposure Improves Eligibility but Does Not Guarantee Safety

Route-partitioned evidence improves completion eligibility. In `policy_docset_v1`, `code_repo_v1`, and `requests`, the broader condition reaches the completion threshold and the controller outputs SAFE. These cases show that the geometry is not merely a refusal rule; when the evidence condition is broad and repair reveals no residual evidence, the controller

can accept the stop claim.

The `urllib3` external audit is the critical boundary case. Route-partitioned evidence has broad support (0.800) and lower localization than the homogeneous condition, but bounded-oracle recall is 0.835, below the completion threshold. The controller outputs CONTINUE, not SAFE. An extended audit reaches support 1.000, recall 1.000, and receives SAFE. This supports the intended claim: broad exposure is completion eligibility, not a sufficient condition for safety.

7.3 Repair Evidence Is Positive but Bounded

Table 2 summarizes the repair boundary. Residual-potential finds residual evidence in several tasks and is useful as a repair instance. On `code_repo_v1`, it recovers more new true items than random and high-potential in the source-route setting. On `urllib3`, it recovers more total new evidence than high-

Table 1: Cross-task evidence-condition diagnostic. Homogeneous route reuse produces localized exposure and would be a false certification if accepted as a global stop in all four tasks. Broader source-route exposure improves completion eligibility. The `urllib3` row is the boundary case: route-partitioned exposure is broad but still incomplete, so the controller outputs CONTINUE rather than SAFE.

Task	Base support	Base Gini	Base recall	False cert.	Base decision	Broad support	Broad recall	Broad decision
<code>policy_docset_v1</code>	0.250	0.771	0.708	True	CONTINUE	0.750	1.000	SAFE
<code>code_repo_v1</code>	0.333	0.750	0.300	True	CONTINUE	1.000	0.950	SAFE
<code>requests</code>	0.250	0.889	0.104	True	CONTINUE	1.000	1.000	SAFE
<code>urllib3</code>	0.200	0.915	0.193	True	CONTINUE	0.800	0.835	CONTINUE

potential. However, the method claim must remain bounded. On `requests`, residual-potential and high-potential select identical target sets and obtain identical gain. On `urllib3`, high-potential is similar or slightly better in cost-normalized evidence despite lower total gain. The main conclusion does not depend on proving that the product rule is uniquely best.

8 Limitations and Additional Validation

The current evidence supports a diagnostic and controller in bounded completion-audit subclasses. It does not establish a universal completion certificate for arbitrary agent workflows. Three limitations are especially important.

First, the external oracles are pattern-defined rather than human annotated. They test whether the mechanism survives real repository structure, but they do not replace a manually curated completion-audit benchmark. Second, the numeric support and Gini thresholds are operational test points, not theory laws. Third, source-route strata are task-designed. Different domains may require different source and route definitions.

The most useful next experiment is not a larger version of the same table, but a claim-verification pilot. In such a task, a workflow receives natural-language claims over a document collection or repository and must certify whether each claim is supported, contradicted, or unresolved. Source-route strata would pair evidence regions with routes such as support search, contradiction search, exception search, temporal qualification, and scope-boundary audit. This pilot would test whether the same certificate mismatch appears when the target is not item recall but claim support. Until it is run, it should be treated as future validation rather than part of the current main claim.

9 Conclusion

This paper studies completion decisions in dynamic agent workflows under unknown workload. The central observation is simple but consequential: local evidence does not automatically support a global completion claim. Stop signals such as no-new findings, agreement, or stable summaries are conditioned on where and how the workflow searched.

Source-route evidence-condition geometry makes this conditioning explicit. The evidence-condition controller uses that

geometry to reject unsafe local stops, repair weak plausible strata, and reserve SAFE for cases where the evidence condition is broad and no residual evidence appears. Across generated and external bounded audits, homogeneous route reuse consistently creates false certification risk, while broader source-route exposure improves completion eligibility without being sufficient by itself. The main contribution is therefore not a claim that a particular repair heuristic is optimal. It is a diagnostic and control principle: completion certificates should be judged by the evidence condition that produced them.

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Table 2: Residual-potential as a bounded repair instance. Residual-potential often finds residual evidence in weak source-route regions, but the evidence does not establish optimality.

Task	Residual gain	High-potential gain	Random gain	Residual/cost	High-potential/cost	Overlap
policy_docset_v1	4.000	0.000	2.025	0.125	0.000	n/a
code_repo_v1	9.000	5.000	4.315	0.136	0.076	n/a
requests	177.000	177.000	45.535	0.054	0.054	1.000
urllib3	329.000	275.000	92.525	0.062	0.063	0.667

Table 3: Leakage control for controller and repair decisions. Runtime decisions use only visible evidence. Oracle labels, oracle totals, post-hoc recall, and undiscovered true item counts are used only after trajectories and challenger choices are fixed.

Component	Runtime allowed	Oracle forbidden
Exposure	visits, scans, actions	missing mass
Support/Gini	exposure counts	post-hoc recall
Potential	text, routes, matches	hidden true items
Decision	condition and repair	target distribution

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Reproducibility Checklist (Draft)

This draft includes computational experiments. The main paper states the evaluable task families, condition types, controller metrics, and leakage controls. A submission-ready version should add the final code archive, environment details, seed policy, exact threshold settings, and scripts that reproduce Tables 1–3. The current experiments use bounded generated tasks and pattern-defined external repository audits; the paper explicitly marks human-annotated completion audits and claim-verification pilots as future validation.